



## Assignment: DSTM

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### Introduction

In today's data-driven society, organizations depend on efficient data management systems to store, process, and analyze vast amounts of information. Understanding how different data types and storage models interact is essential for designing reliable and scalable solutions.

This report explores key concepts in modern data management, including the differences between structured, semi-structured, and unstructured data, and their relationship to SQL and NoSQL systems. It also examines the ACID, BASE, and CAP principles, compares databases, data warehouses, data marts, and data lakes, and discusses the role of metadata in ensuring structure and governance. Finally, the Inmon–Kimball debate is presented to illustrate contrasting approaches to data warehouse design.

### 1. Method For Literature Research

The literature research was conducted through Dalarna University library's Summon discovery tool, which indexes a wide range of databases accessible via the university (Dalarna University library, n.d.).

These databases were selected because they cover disciplines relevant to the study of data science, database management systems and information technology, ensuring that the retrieved material represents credible and academically reviewed sources. In addition to academic sources, a few relevant articles from the internet were also advised to complement the findings.

Table 1. Literature search process conducted through the Dalarna University library.

| Search Step | Keywords / Search String                                                                     | Database / Search Engine   | No. of Results | Comments / Notes                                 |
|-------------|----------------------------------------------------------------------------------------------|----------------------------|----------------|--------------------------------------------------|
| 1           | "types of data" AND databases                                                                | Dalarna University Library | 10,610         | Broad overview of how data types are categorized |
| 2           | "structured data" OR "unstructured data" OR "semi-structured data"                           | Summon                     | 25,574         | Get academic definitions of data types           |
| 3           | "structured data" AND "relational database"                                                  | Summon                     | 519            | Start connecting data type to DBMS               |
| 4           | "structured data" OR "semi-structured data" OR "unstructured data" AND "relational database" | Dalarna University Library | 719            | Connecting all data types to DBMS                |

## 1.1 Structured, Semi-structured and Unstructured Data and their relationship to different types of DBMS

### 1.1.1 Definitions and characteristics

Data in modern information systems is generally categorized as structured, semi-structured or unstructured, depending on the level of schema organization and data regularity. Structured data refers to information organized in fixed fields following a strict schema, typically represented as rows and columns in a relational database (Sambrekar et al., 2018). This data type can be easily stored, queried and analyzed using SQL-based systems such as MySQL, PostgreSQL or Oracle. Examples include financial transactions, employee records and inventory logs (IBM, n.d.-a).

Semi-structured data lacks a rigid schema but maintains self-describing structure through metadata, tags or delimiters (Sambrekar et al., 2018). Formats such as JSON, XML or YAML exemplify this category, providing flexibility in its schema model. Semi-structured data bridges the gap between the highly structured nature of relational data and the flexibility demanded by modern web and IoT applications.

Unstructured data represents information without predefined data models or consistent formats. It includes free text, social media content, multimedia files, etc. (Sambrekar et al., 2018). Such data typically originates from diverse sources and requires processing or machine learning techniques for interpretation (Paiva et al., 2025).

### 1.1.2 Relationship Between Data Types and DBMS

The connection between data type and database management approach is central to system design.

SQL-based relational DBMS handle structured data efficiently and follow the ACID principles of atomicity, consistency, isolation, and durability, ensuring strong data integrity in transactional environments (Paiva et al., 2025).

In contrast, NoSQL databases, including document, key-value, column-family, and graph models, support semi-structured and unstructured data through flexible schemas. They follow the BASE model (Basically Available, Soft state, Eventual consistency), prioritizing scalability and availability over strict consistency (Sambrekar et al., 2018; MongoDB Inc., n.d.).

Studies show that document-based systems like Couchbase and MongoDB outperform traditional SQL databases in scalability and insert operations for unstructured data (Sambrekar et al., 2018). Modern systems increasingly adopt hybrid architectures, where relational databases manage structured data and NoSQL systems handle high-volume or variable content (Paiva et al., 2025).

## 2. Method For Literature Research

The literature review was conducted through Dalarna University Library's Summon database (Dalarna University Library, n.d.), focusing on peer-reviewed publications explaining and comparing the ACID, BASE, and CAP principles of database management. Searches combined terms such as "BASE analysis of database" and "CAP theorem database consistency", later refined to studies on their application in

distributed and NoSQL systems. To complement academic findings, accessible sources from Medium (Vikram, n.d.) and Towards Data Science (Lang, 2020) were also included for practical context.

## 2.1 Literature search process carried out via Dalarna University's library's database.

| Search Step                | Keywords / Search String                      | Database / Search Engine                       | No. of Hits                                 | How / Selection                                                              |
|----------------------------|-----------------------------------------------|------------------------------------------------|---------------------------------------------|------------------------------------------------------------------------------|
| Theoretical in-depth study | "ACID components database systems"            | Dalarna University Library / Summon            | Several relevant                            | Articles explaining ACID properties and application in relational databases. |
| Concept/ Explanations      | "ACID transactions explained"                 | Google Scholar / Medium / Towards Data Science | Multiple hits, selection based on relevance | Vikram (n.d.) and Lang (2020), clear and practical explanations of concepts. |
| Extended search            | "Big data storage technologies ACID BASE CAP" | Google Scholar                                 | Limited quantity (overview articles)        | Selection: Siddiqua et al. (2017), compare ACID, BASE and                    |

## 3.2 Significance of ACID, BASE, and CAP Principles in Database Systems

The evolution of databases, from early relational systems to today's distributed and Big Data platforms, has shifted focus from pure stability to balancing consistency, availability, and performance.

The ACID, BASE, and CAP principles represent three complementary philosophies. ACID ensures reliability and data integrity, BASE prioritizes flexibility and availability and CAP describes the theoretical limits preventing systems from achieving all three properties simultaneously (Deka, 2015; Siddiqua et al., 2017; Anagnostakis & Glavas, 2025; Naseri Seyedi Noudoust et al., 2022).

Each model applies to different contexts: ACID remains vital for systems demanding accuracy, BASE suits large-scale distributed environments, and CAP explains the trade-offs among consistency, availability, and partition tolerance (Siddiqua et al., 2017; Deka, 2015).

In practice, many modern systems adopt hybrid architectures that combine ACID's stability, BASE's scalability, and CAP's theoretical insight, creating databases that meet the dual demands of precision and performance in data-driven applications (Naseri Seyedi Noudoust et al., 2022; Vikram, n.d.; Lang, 2020).

## 3. Differences between Database, Data Warehouse and Data Lake

Data storage technologies form the foundation of modern information systems and analytical processes. As data volumes and variety have grown, so too have the methods for managing, processing and interpreting information. Understanding the underlying structures, principles and architectures of data systems is essential for effective decision making in organizations that are more data driven. The following sections explore key concepts within data science and database management, including the differences between structured, semi-structured and unstructured data, the significance of ACID, BASE

and CAP principles, the distinction between databases, data warehouses, data marts and data lakes, the role of metadata and comparison of the influential approaches proposed by Bill Inmon and Ralph Kimball.

Table 2. Literature search process conducted through the Dalarna University library.

| Search Step | Keywords / Search String                                               | Database / Search Engine   | No. of Results | Comments / Notes                                                                                           |
|-------------|------------------------------------------------------------------------|----------------------------|----------------|------------------------------------------------------------------------------------------------------------|
| 1           | "database" OR "data storage systems"                                   | Dalarna University Library | 23,388,553     | Broad overview of database architectures                                                                   |
| 2           | "data warehouse" OR "data warehousing"                                 | Summon                     | 430            | Collected general background literature on data-warehouse principles and definitions                       |
| 3           | "Building the Data Warehouse" AND Inmon / "The Data Warehouse Toolkit" | Summon                     | 10 / 29        | Located <i>Building the Data Warehouse</i> (Inmon) and <i>The Data Warehouse Toolkit</i> (Kimball & Ross). |

### 3.1.2 Databases

A database is a structured system designed for online transaction processing (OLTP). It stores current operational data such as orders or payments in normalized tables to ensure accuracy and integrity (Inmon, 2005). Relational databases use SQL and follow ACID principles, guaranteeing consistency and reliability. Examples include MySQL, PostgreSQL and Oracle, which excel in real-time updates but are less suited for large-scale analytical queries (Kimball & Ross, 2013).

### 3.1.3 Data Warehouses

A data warehouse (DW) is a centralized repository for analytical processing rather than transactions. Inmon (2005) defines it as subject-oriented, integrated, time-variant, and non-volatile, consolidating data from multiple sources into a consistent analytical structure. Data warehouses store structured or semi-structured data using ETL or ELT processes and are optimized for OLAP, enabling historical analysis and reporting. They ensure a single version of truth across the enterprise.

### 3.1.4 Data Marts

A data mart is a smaller, department-focused subset of a data warehouse, supporting specific business areas such as sales or marketing (Kimball & Ross, 2013). Kimball's bottom-up approach treats data marts as building blocks connected through a bus architecture of conformed dimensions, while Inmon views them as dependent on a central warehouse. Both agree data marts handle structured data optimized for OLAP queries.

### 3.1.5 Data Lakes

A data lake stores raw structured, semi-structured, and unstructured data in its original form, allowing scalability and flexibility (Dixon, 2019; IBM, n.d.-b). It applies schema-on-read, adding structure only when data is accessed. Data lakes support machine learning and data discovery, complementing warehouses by accommodating diverse data sources such as JSON logs, IoT streams, and multimedia files.

## 4. The Role of Metadata in Data Warehouses and Data Lakes

In modern data-driven organizations, large amounts of information are collected, processed and stored across different platforms. As these data environments grow and complexity increase, it becomes increasingly difficult to understand what the information represents, where it comes from, and how it should be used. Here, metadata, often described as "data about data", is considered a crucial role. Metadata acts as a bridge between raw data and meaningful information. This enables users, analysts and systems to understand the structure, context, and quality of data stored in both Data Warehouses and Data Lakes.

While a data software typically handles structured data organized into predefined schemas, a data lake stores raw, semi-structured and unstructured data from many different "data swamps" (Sawadogo & Darmont, 2021). According to Nambiar and Mundra (2022), metadata plays a central role not only in describing data, but also in supporting data integration, governance, and decision-making.

This section explores four key questions that together explain the importance of metadata in data storage environments:

1. Do we need metadata?
2. Are there different types of metadata?
3. How do we generate metadata?
4. How do we store metadata?

Through these questions, they compare how metadata is used, created, and maintained in data warehouses and data lakes, highlighting their similarities and differences in today's data-driven environments.

### 4.2 Literature search process carried out using Dalarna University's library database.

| Steps for search        | Keywords/Search String                       | Database / Search Engine    | Applied filters         | Number of hits |
|-------------------------|----------------------------------------------|-----------------------------|-------------------------|----------------|
| 1. Broad search         | "metadata AND data Warehouses"               | Summon (Dalarna University) | Peer-reviewed, fulltext | Approx: 236    |
| 2. First delimitation   | "metadata AND data Warehouses OR Data Lakes" | Summon (Dalarna University) | Peer-reviewed, fulltext | Approx: 182    |
| 3. Name-specific search | data warehouses OR data lakes                | Summon (Dalarna University) | Peer-reviewed, fulltext | Approx: 2,607  |

#### 4.2.1 The Importance of Metadata

Metadata, often described as data about data, is essential for understanding, organizing, and governing information. It defines how data is structured, where it originates, and how it can be used. Without metadata, organizations would have large volumes of data but little ability to interpret or ensure its quality.

In data warehouses (DW), metadata provides control and structure by defining schemas before data is stored, ensuring consistency and reliability (Wu, Miller, & Nilakanta, 2001). In contrast, data lakes (DL) deal with raw and often unstructured information, where metadata serves as a map that enables discovery, context, and governance. Without it, the data lake risks turning into a data swamp, where data lacks meaning and usability (Sawadogo & Darmont, 2021).

Metadata typically contains details such as file format, source, load time, ownership, and data quality. In short, data warehouses use metadata to ensure structure and order, while data lakes use it to provide understanding and searchability (Nambiar & Mundra, 2022).

#### 4.2.2 Types of Metadata

Both environments rely on three main categories of metadata (Wu et al., 2001):

- Technical metadata – describes schemas, data types, and lineage.
- Business metadata – defines meaning, ownership, and business rules.
- Process metadata – records data movement, updates, and quality checks.

While these categories exist in both systems, data warehouses emphasize control and standardization, whereas data lakes focus on classification and semantics (Sawadogo & Darmont, 2021; Moges et al., 2016).

#### 4.2.3 Metadata Generation

Metadata can be manual, automatic, or hybrid.

- Manual generation involves documenting business rules, ownership, and KPIs to ensure consistency.
- Automatic generation occurs during ETL or ELT processes, where systems extract technical metadata about tables, relationships, and transformations (Wu et al., 2001).
- Hybrid generation combines both, ensuring that metadata is technically accurate and semantically meaningful (Moges, Van Vlasselaer, & Lemahieu, 2016).

In data warehouses, metadata is mainly generated automatically during ETL flows. In data lakes, catalog services such as AWS Glue or Apache Atlas extract metadata during ingestion, while users add business context manually (Eichler, Giebler, & Gröger, 2021; Sawadogo & Darmont, 2021).

#### 4.2.4 Metadata Storage

In data warehouses, metadata is typically held in a centralized repository integrated with ETL tools, documenting technical and business details (Wu et al., 2001). This ensures consistency and reliability in structured environments.

Data lakes, however, use distributed catalogs (e.g., AWS Glue, Google Cloud Data Catalog) that index files externally for flexibility and scalability (Eichler et al., 2021). These systems must be continuously updated to avoid the data swamp problem (Sawadogo & Darmont, 2021).

Ultimately, metadata storage connects data to meaning. Inmon and Kimball's principles apply here as well: structure ensures control, while flexibility supports discovery. Well-managed metadata allows organizations not only to know what data they have, but also to trust and use it confidently (Nambiar & Mundra, 2022).

### 5. The Inmon–Kimball Debate

The design and implementation of data warehouses have been widely influenced by two main thinkers, Bill Inmon and Ralph Kimball. Their contrasting philosophies, often referred to as the Inmon-Kimball Debate, continue to shape how organizations choose to structure, integrate and deliver analytical data. While both aim to achieve reliable, consistent and accessible business information, they differ in architectural focus, data modeling and delivery methods.

#### 5.1.1 The Inmon Approach: Top-down

Bill Inmon's philosophy represents a top-down approach, beginning with the creation of centralized Enterprise Data Warehouse (EDW) that integrates data from multiple operational systems. He defines a data warehouse as subject-oriented, integrated, non-volatile and time-variant (Inmon, 2005). This means the warehouse is designed around business subjects rather than applications, integrates data from various sources, stores historical snapshots and maintains data that is stable over time.

Inmon's model prioritizes data consistency and governance. Data is extracted from operational databases, cleansed, standardized and loaded into a 3rd Normal Form (3NF) repository, ensuring that enterprise data definitions are unified before analysis. From this central warehouse, dependent data marts can be created for specific departments for analytical needs (Inmon, 2005).

A crucial concept in Inmon's design is granularity, referring to the level of detail stored within the warehouse. Determining granularity affects both performance and storage efficiency. According to Inmon (2005), higher granularity (more detail) increases analytical flexibility but also requires more storage and careful management.

Overall, Inmon's approach aims to create a single version of the truth, ensuring enterprise-wide consistency, but often at the cost of longer development time.

## 5.2 The Kimball Approach: Bottom-up

Ralph Kimball proposes a bottom-up approach, where the data warehouse is built incrementally through integrated data marts. Each mart focuses on a specific business process, such as sales or inventory and uses a dimensional model, commonly organized into fact and dimension tables following the star schema design (Kimball & Ross, 2013).

Kimball's architecture uses what he calls the bus architecture, where conformed dimensions (such as time, customer or product) ensure that all data marts can be connected consistently across the enterprise. Integration therefore occurs through these shared documents rather than through a central 3NF model.

The dimensional modelling process consists of four key steps:

1. Selecting the business process
2. Declaring the grain level (level of detail)
3. Identifying the dimensions
4. Identifying the facts (Kimball & Ross, 2013)

Kimball's approach is often considered more agile and business-oriented, since each data mart can deliver immediate analytical value while still fitting an integrated enterprise framework. It also prioritizes usability and performance, as dimensional models are easier for analysts to query compared to complex normalized schemas.

In essence, both approaches seek the same goal, providing high quality, accessible analytical data, but differ in how they prioritize structure versus speed.

### 5.2.1 Structural and Methodological Differences

The main difference between the two approaches lies in how integration and modelling are achieved.

- Inmon's Corporate Information Factory (CIF) uses a normalized 3NF EDW as the foundation, ensuring integration before users access the data.
- Kimball, meanwhile, enforces integration through conformed dimensions at the presentation layer, allowing multiple data marts to function as a unified analytical environment (Kimball & Ross, 2013).

### 5.2.2 Data Marts, End-user access and Usability

Both Inmon and Kimball discuss data marts, but they assign them different roles.

- In Inmon's architecture, data marts are dependent, created only after the EDW is established and populated with verified data (Inmon, 2005). This maintains accuracy and enterprise alignment.
- In Kimball's model, data marts are independent building blocks that form the foundation of the warehouse. They are designed around business needs and later integrated through conformed dimensions (Kimball & Ross, 2013).



From an end-user perspective, Inmon's normalized model is not easily able to be queried by business users, requiring separate analytical databases or marts for accessibility. Kimball's dimensional approach, on the other hand, is intuitive and performance optimized, designed for direct user interaction and simplified reporting.

## Conclusion

Modern data management depends on understanding how different data types, structured, semi-structured, and unstructured, shape the way information is stored and processed. Structured data fits well within relational databases, where schema and integrity are prioritized, while semi-structured and unstructured data are better managed through NoSQL systems and data lakes, which offer flexibility and scalability.

This report has demonstrated how the ACID, BASE, and CAP principles define trade-offs in consistency and performance, and how platforms such as data warehouses and data marts support analytical insight. It also emphasized the role of metadata in ensuring governance and trust in data systems. Finally, the comparison between Inmon and Kimball highlighted how differing architectural philosophies can together guide the design of efficient, reliable, and adaptable data environments.

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